ML Lab Week 14: CNN Image Classification

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1. Introduction

The objective of this lab was to design, implement, and train a Convolutional Neural Network (CNN) capable of classifying images of hand gestures into the categories rock, paper, and scissors. The task involved preparing the dataset, constructing the model architecture, training it using supervised learning, and evaluating its performance on unseen test images.

2. Model Architecture

• Describe the CNN architecture you built.

The CNN consists of three convolutional blocks with increasing channel sizes, each followed by a ReLU activation and a MaxPool layer to progressively extract spatial and hierarchical features from the input images.

• Mention key parameters like kernel size, number of channels, and the use of Max Pooling.

Each convolution layer uses a 3×3 kernel with padding 1, and the channel depth increases from 16 to 32 to 64. Every block includes a MaxPool2d layer with a 2×2 kernel to reduce spatial dimensions by half and improve feature abstraction.

• Describe the fully-connected classifier.

The flattened feature map (64×16×16) is passed into a fully connected network consisting of a 256-unit Linear layer with ReLU activation and dropout (p=0.3), followed by a final Linear layer that outputs scores for the three gesture classes.

3. Training and Performance

• State the key hyperparameters used for training: optimizer, loss function, learning rate, and number of epochs.

The model was trained using the Adam optimizer with a learning rate of 0.001, CrossEntropyLoss as the loss function, and a total of 10 training epochs.

• Report the final Test Accuracy your model achieved.

The final test accuracy achieved by the model was **98.40%.**

4. Conclusion and Analysis

• Briefly discuss your results. Did the model perform well?

Yes, the model performed extremely well, achieving high accuracy and demonstrating strong ability to classify unseen gesture images.

• Were there any challenges you faced?

The main challenges involved ensuring correct preprocessing steps, managing input dimensions for the fully connected layer, and verifying proper dataset splitting and device selection.

• Suggest one or two ways you could potentially improve the model's accuracy in the future.

Accuracy could be improved by introducing data augmentation techniques (such as rotation or flipping) or by adding Batch Normalization or deeper convolutional layers to enhance generalization.